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[23TBSCSMA31]

THIRD SEMESTER B.Tech EXAMINATION, SEPTEMBER-2024

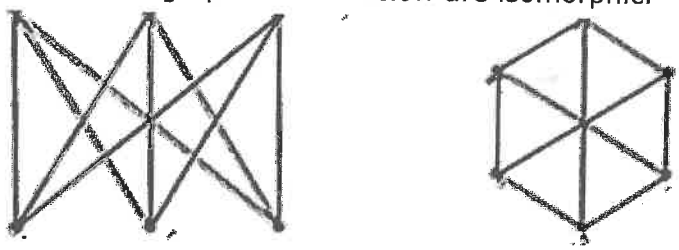
DISCRETE MATHEMATICS AND GRAPH THEORY

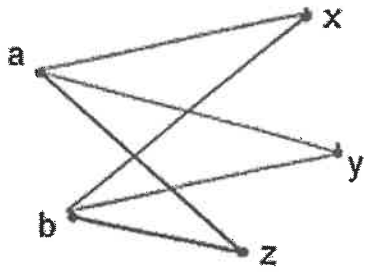
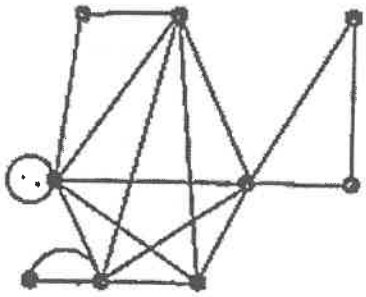
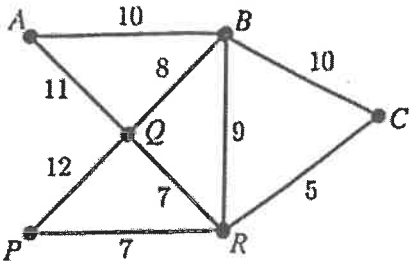
Time: 3 Hours **Maximum Marks: 100**

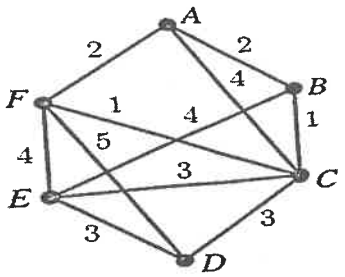
Instructions: i) **Missing data may be suitably assumed**

ii) **Answer all Questions**

		PART-A	5 x2=10		
1.	a.	Cars of a particular manufacturer come in 4 models, 12 colours, 3 engine sizes, and 2 transmission types. i. How many distinct cars (of this company) can be manufactured? ii. How many of these have the same colour?	L-2	CO1	02
	b.	Find the sequence generated by the $(3+x)^3$.	L-2	CO1	02
	c.	Solve the recurrence relation $a_n = 7a_{n-1}$, where $n \geq 1$, given that $a_2 = 98$.	L-2	CO1	02
	d.	Define null graph and multi graph.	L-1	CO1	02
	e.	State balanced tree with an example.	L-1	CO1	02
		PART-B	**6 X 5=30**		
2.	a.	How many friends must you have to guarantee that at least 5 of them will have birthdays in the same month?	L-3	CO2	05
		OR			
	b.	Find the number of integer solutions of $x_1 + x_2 + x_3 + x_4 + x_5 = 30$ Where $x_1 \geq 2,\ x_2 \geq 3,\ x_3 \geq 4,\ x_4 \geq 2\ and\ x_5 \geq 0$.	L-3	CO2	05
3.	a.	Determine the coefficient of x^{15} in $\frac{(1+x)^4}{(1-x)^4}$.	L-3	CO3	05
		OR			
	b.	Using generating function, find the number of non-negative integer solutions of the equation $x_1 + x_2 + x_3 + x_4 = 25$.	L-3	CO3	05
4.	a.	Solve the recurrence relation $a_n = 6a_{n-1} - 9a_{n-2}$ with the initial conditions $a_0 = 1$ and $a_1 = 4$.	L-3	CO4	05
		OR			
	b.	There are 3 pegs fixed vertically on a table, and n circular disks having holes at their centers and having increasing diameters are slipped onto one of these pegs, with the largest disk at the bottom. The disks are to be transferred, one at a time, onto another peg with the condition that at no time a larger disk is put on a smaller disk. Determine the number of moves for the transfer of all the n disks, so that at the end the disks are in their original order.	L-3	CO4	05
5.	a.	Prove that in every graph, the number of vertices of odd degrees is even.	L-2	CO5	05
		OR			

	b.	Verify that the two graphs shown below are isomorphic. 	L-2	CO5	**05**
6.	**a.**	A computer laboratory of a school has 10 computers that are to be connected to the wall socket that has two outlets. The connections are made by using extension cords that have two outlets each. Find the least number of cords needed to get these computer setup for use.	L-3	CO6	**05**
		OR			
	b.	Using the merge sorting method sort the following list 38, 27, 43, 3, 9, 82, 10	L-3	CO6	**05**
7.	**a.**	Seven books are distributed among 7 students for reading. The books are recollected and redistributed. In how many ways will each student get to read two different books	L-3	CO2	**05**
		OR			
	b.	Prove that the number of partitions of a positive integer n into distinct summands is equal to the number of partitions of n into odd summands.	L-3	CO3	**05**
		PART-C	**6X10=60**		
8.	**a.**	Obtain the rook polynomial for the 3×3 board by using expansion formula.	L-3	CO2	**10**
		OR			
	b.	How many ways can 26 letters of the English alphabets be permuted so that none of the patterns **CAR, DOG, PUN**, or **BYTE** occurs?	L-3	CO2	**10**
9.	**a.**	Using generating function, find the number of partitions of $n = 6$ and write all possible partitions of $n = 6$.	L-3	CO3	**10**
		OR			
	b.	Using exponential generating function, find the number of ways in which 4, 5 and 6 of the letters in the word ***INLINE*** be arranged.	L-3	CO3	**10**
10.	**a.**	Solve the recurrence relation $a_n - a_{n-1} - 2a_{n-2} = 4$, given $a_0 = 0$ and $a_1 = -1$	L-3	CO4	**10**
		OR			
	b.	Solve the recurrence relation $a_{n+1} - a_n = n^2,\ n \geq 0$, with $a_0 = 1$ using generating functions.	L-3	CO4	**10**

11.	**a.**	If λ denotes the number of colors available to color the vertices of the graph $K_{2,3}$ given below.  Find: **i.** how many proper colorings of the graph have vertices a, b colored the same. **ii.** how many proper colorings of the graph have vertices a, b colored with different colors. **iii.** the chromatic polynomial of the graph. **iv.** the chromatic polynomial for the graph $K_{2,n}$.	L-3	CO5	**10**
		OR			
	b.	Carry out the elementary reduction process for the following graph shown below and check the planarity. 	L-3	CO5	**10**
12.	**a.**	Explain the following with an example. **i.** Walk **ii.** Trail **iii.** Path **iv.** Circuit **v.** Cycle	L-2	CO1	**10**
		OR			
	b.	Using Kruskal's Algorithm, find a minimal spanning tree for the given weighted graph.  	L-3	CO6	**10**
13.	**a.**	Obtain an optimal prefix code for the message ROAD IS GOOD. Indicate the code.	L-3	CO6	**10**

		OR			
	b.	Using Prim's algorithm, find a minimal spanning tree of the weighted graph. 	L-3	CO6	**10**